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The Impact of Population Ageing on Public Finances in the EU-27

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ABSTRACT

This paper analyses the impact of demographic ageing on age-related public expenditure in the EU-27 from 1992 to 2024. The analysis covers healthcare, long-term care (LTC), and pensions. The results reveal three patterns. First, demographic effects are heterogeneous across countries, driven by short-run responses that vary by spending category: limited in healthcare, absent in LTC. Second, advanced ageing (80+) matters more for long-run dynamics, especially in healthcare. Third, Central and Eastern European economies show weaker or delayed responses, while Western and Nordic countries adjust more strongly, especially in healthcare and LTC rather than pensions, reflecting the decisive role of institutional design.

Keywords: Ageing, public expenditure, demographic change, European Union

JEL classification: E60, E62, H20, H51, H55**,** J11, J14

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1. INTRODUCTION

Population ageing is a principal driver of public finance dynamics in advanced economies. The combination of persistently low fertility rates, sustained gains in life expectancy, and weak demographic growth, insufficiently offset by migration, has led to a steady increase in the share of individuals aged 65 and over in the working-age population, that is, the ratio between the population aged 65+ and those aged 15-64. This demographic transformation has become a central concern for policymakers and researchers alike, given its potential implications for the sustainability of public finances and welfare systems.

The fiscal consequences of ageing operate through multiple channels. On the expenditure side, it increases demand for pensions, healthcare services, and long-term care. On the revenue side, demographic change tends to reduce labor-force participation and weaken labor-based tax bases, thereby affecting governments' tax revenues. The issue is particularly relevant in the EU, where member states face broadly similar demographic trends. However, countries differ in their institutional arrangements, welfare-state models, and public policy frameworks, which may shape how demographic pressures translate into public expenditure. Consequently, similar demographic developments may generate markedly different outcomes.

Understanding how demographic change translates into public expenditure pressures across different institutional settings is essential for assessing the long-term implications of ageing and designing effective policy responses. A substantial body of literature has examined this relationship, generally finding that ageing increases pressure on public expenditure, particularly through pensions, healthcare, and long-term care. However, most studies focus on individual expenditure categories, country case studies, or limited groups of countries, making it difficult to determine whether the fiscal consequences of ageing follow common patterns across Europe or are largely shaped by national institutional characteristics.

Against this background, this paper pursues three main objectives. First, it examines the relationship between population ageing and age-related public expenditure across the EU-27 from 1992 to 2024, covering healthcare, long-term care, and pensions. Second, it looks for cross-country heterogeneity in expenditure responses to demographic change. Third, it assesses differences between extensive and intensive ageing and distinguishes short-run from long-run effects of ageing in the relationship. To this end, our empirical strategy combines dynamic Error Correction Models (ECM), mixed-effects estimations, and cluster analysis, providing new evidence on the heterogeneous fiscal implications of population ageing across European countries.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature. Section 3 presents the data, variables and describes the econometric methodology. Section 4 discusses the empirical results. Finally, section 5 concludes and outlines the main policy implications.

2. LITERATURE REVIEW

Population ageing has become a central structural determinant of fiscal dynamics in advanced economies. A broad empirical consensus identifies demographic change as a

key driver of long-term fiscal imbalances, although the magnitude and transmission mechanisms of these effects remain highly heterogeneous across countries and methodological approaches (OECD, 2022a; European Commission, 2024; IMF, 2023).

The theoretical foundations of this literature are primarily grounded in life-cycle theory and overlapping-generations (OLG) models (Auerbach et al., 1994; Auerbach and Kotlikoff, 1999), which formalise the intertemporal distribution of fiscal contributions and transfers across cohorts. In these frameworks, individuals are net contributors during working age and net beneficiaries in retirement, implying that increases in old-age dependency ratios generate a structural deterioration in fiscal balances. OLG models also emphasise broader macroeconomic channels, including changes in savings behaviour, labour supply, productivity, and public debt dynamics under intertemporal budget constraints (Beetsma et al., 2025; Ramos-Herrera and Sosvilla-Rivero, 2020).

Empirical research identifies two primary transmission channels. On the revenue side, population ageing erodes the labour tax base by reducing the working-age population, lowering labour-force participation, and moderating wage growth. Given the centrality of labour taxation in OECD and EU fiscal systems, this mechanism structurally weakens revenue capacity. A substantial body of panel-data evidence documents a negative association between ageing indicators and fiscal revenues or overall budget balances (Filipović & Miljković, 2024; OECD, 2022a), although estimated effects vary across institutional contexts. In contrast, consumption and capital taxation appear less directly affected, reflecting differences in behavioural responses, tax design, and the relatively low taxation of wealth (Benzarti and Carloni, 2019; Piketty and Saez, 2013; OECD, 2021).

On the expenditure side, ageing is consistently associated with rising public spending, particularly on pensions, healthcare, and long-term care. Pension expenditure shows the most direct and robust link to demographic structure, especially in pay-as-you-go systems where fiscal sustainability depends critically on the contributor-beneficiary ratio (Verbič & Spruk, 2024). Empirical evidence confirms a positive elasticity between ageing indicators and pension spending, although parametric reforms, such as increases in statutory retirement ages and adjustments to benefit formulas, have partially mitigated these pressures (Aguilar-Segado, 2024; Szwarc, 2024). Healthcare and long-term care expenditures are also expected to increase, driven by higher prevalence of chronic conditions, end-of-life costs, and growing demand for care services, although their dynamics are shaped by non-demographic factors such as technological progress, institutional design, and efficiency (Cylus et al., 2022; OECD, 2024; Navarro-García and Sarria-Santamera, 2023).

Beyond these direct fiscal effects, a growing strand of literature highlights indirect macroeconomic channels through which ageing affects fiscal sustainability (Ramos-Herrera and Sosvilla-Rivero, 2020; OECD, 2022a; Filipović and Miljković, 2024). Demographic change is associated with lower potential output because it reduces labour supply, slows productivity growth, and shifts sectoral composition toward less productive activities (Aksoy et al., 2019; Beetsma et al., 2025). These dynamics weaken fiscal capacity by reducing both the tax base and economic growth. However, the net effect remains ambiguous, as labour scarcity may induce capital deepening, automation, and technological adoption, partially offsetting productivity losses (Acemoglu and Restrepo, 2022).

A central insight of the empirical literature is the substantial heterogeneity in the fiscal effects of ageing. Cross-country differences in pension systems, labour-market institutions, tax structures, and healthcare organisation significantly mediate the impact of demographic change (Brys et al., 2016; Petrucci and Margarito, 2025; Cadaval-Sampedro et al., 2026). As a result, similar demographic trends can produce markedly different fiscal outcomes depending on institutional configurations. Empirical findings reflect this heterogeneity, ranging from strongly negative effects to statistically insignificant relationships once macroeconomic controls and institutional variables are included (Twarowska and Twarowska-Mól, 2024).

Methodological challenges further complicate empirical identification. Many studies rely on aggregate demographic indicators that fail to capture heterogeneity in health status, effective labour supply, or cohort composition. Moreover, endogeneity concerns are pervasive, as fiscal variables and demographic dynamics are jointly determined through feedback mechanisms, including those linking public expenditure, health outcomes, and longevity (Ajovalasit et al., 2024; Iuga et al., 2024). While recent contributions employ dynamic panel models, system-GMM estimators, and stochastic projections, causal identification remains a central challenge.

Recent research has also incorporated additional moderating factors. Migration and increased labour-force participation may partially offset demographic pressures, although their long-term impact appears limited (OECD, 2020; Michalski and Stępień, 2021). Improvements in population health may delay retirement and reduce expenditure pressures (Iuga et al., 2024), while political economy constraints, particularly the increasing electoral weight of older cohorts, may hinder the implementation of fiscal reforms (Brosig, 2022; Barilari et al., 2024).

3. EMPIRICAL STRATEGY

The analysis focuses on the three main public expenditure categories related to ageing: healthcare (h), long-term care (ltc), and pensions (p), as well as their aggregate[1] ($a = h + ltc + p$). All of them are expressed as a percentage of GDP. We use an unbalanced panel dataset covering 27 European Union countries. The main data source is Eurostat, which ensures consistent definitions and comparability across countries and over time. The overall dataset covers 1992–2024, but the expenditure series have different temporal coverage. Healthcare and long-term care data extend to 2024, whereas pension expenditure is available only through 2023 in the data release used in this study; consequently, the aggregate measure, which includes pensions, also ends in 2023. Coverage also varies across countries, particularly for healthcare and long-term care, producing an unbalanced panel. Appendix E reports the number and temporal coverage of observations available for each country and expenditure category. The effective estimation samples are smaller because the specification requires first differences and

[1] The aggregate measure is calculated only when all three expenditure components are simultaneously available. Eurostat's harmonised definitions enhance comparability across countries and years, although differences in national accounting and institutional arrangements may remain, particularly for long-term care. The variables should therefore be interpreted as harmonised indicators rather than perfectly equivalent measures of national welfare provision.

lagged variables, as well as complete observations for all controls; nevertheless, all models retain the 27 EU countries.

The key regressors capture two distinct demographic dimensions of population ageing. The share of the population aged 65 and over (pc65) measures its extensive dimension, that is, the overall size of the elderly population. By contrast, the share of the population aged 80 and over relative to those aged 65 and over (p80) captures the intensive dimension of ageing by measuring the age composition within the elderly population. This distinction matters because the expected effect of intensive ageing differs across expenditure categories. A higher concentration of individuals aged 80 and over should exert particularly strong upward pressure on healthcare and long-term care expenditure, as the prevalence and severity of health problems, functional limitations, and dependency tend to increase markedly beyond this age threshold. Consequently, conditional on the overall share of the elderly population, a shift in its composition towards the oldest age groups is expected to increase both the probability of requiring care and the intensity of the services received. The relationship may be different for pension expenditure. Holding pc65 constant, a larger p80 implies a greater concentration of pensioners who retired further in the past. Because pension benefits generally uprate less rapidly than wages or professional earnings, their real or relative value tends to decline over time since retirement. The intensive component of ageing may therefore be negatively associated with average pension expenditure and may partly offset the positive expenditure effect arising from an increase in the total elderly population.

The econometric specification also includes a set of standard macroeconomic controls: real GDP growth, GDP per capita in purchasing power standards, total government expenditure as a share of GDP, and the fiscal balance (deficit/surplus). These variables capture cyclical and structural factors that affect public spending dynamics. The last three are lagged by one period to avoid simultaneity bias. We also introduce a dummy variable to capture the COVID-19 period (2020–2021), which is expected to have substantially affected public expenditure relative to GDP. First, because GDP fell in most countries. Second, the required additional healthcare spending. **Table 1** provides detailed descriptions and definitions of all variables.

<Insert Table 1>

To capture both the persistence of public expenditure and the effects of demographic ageing, we model public expenditure as a function of its lagged value and the variables listed above. We combine Error Correction Models (ECM) for Time-Series Cross-Section (TSCS) data and mixed-effects specifications to account for unobserved heterogeneity, dynamic adjustment, and cross-country differences in responses. We use an error-correction parameterization because it is algebraically equivalent to the underlying autoregressive distributed lag model and clearly separates short-run responses, adjustment dynamics, and the long-run effects of the ageing variables, even when the process is stationary but highly persistent. This is particularly useful in TSCS settings,

where the substantive interest often lies in both short-run responses and gradual adjustment over time (De Boef and Keele, 2008; Beck and Katz, 2011).[2]

Even so, this econometric framework does not fully resolve the identification challenges inherent in observational data. Establishing causality would require exogenous sources of variation and stronger identification strategies. Accordingly, our results should best be viewed as providing robust evidence of systematic associations and dynamic relationships, rather than conclusive proof of causal effects.

We estimate the ECM using Least Squares Dummy Variables (LSDV) with country-specific effects, as it offers a parsimonious and interpretable way to model persistence, long-run adjustment, unit-specific intercepts, and heterogeneous effects.[3] The ECM specification is given by:

$$\Delta \exp_{it} = \alpha \exp_{it-1} + \beta_1 pc65_{it-1} + \beta_2 p80_{it-1} + \lambda_1 \Delta pc65_{it} +$$

$$+\lambda_2 \Delta p80_{it} + \delta' \Delta X_{it} + +\delta COVID_t + \mu_i + \varepsilon_{it} \qquad [1]$$

where:

- $\exp_{it}$ denotes public expenditure (healthcare, long-term care, pensions, or total expenditure) expressed as a percentage of GDP
- $pc65_{it}$ is the share of the population aged 65 and over
- $p80_{it}$ is the share of the population aged 80 and over relative to the elderly population
- X_{it} is the vector of macroeconomic controls
- $COVID_t$ is a dummy variable equal to 1 for the years 2020–2021[4]
- μ_i captures unobserved country-specific effects
- ε_{it} is the idiosyncratic error term

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[2] For stationary variables, the long-run multiplier describes the cumulative response to a permanent change in a regressor, subject to dynamic stability and correct specification. As we show below, unit-root tests reject the null for most expenditure categories, although the evidence is less conclusive for one spending category. Given the difficulty of distinguishing a unit root from a highly persistent stationary process, we use a common dynamic parameterization for comparability, report the tests separately by category, and examine robustness to alternative specifications.

[3] The inclusion of a lagged dependent variable in a fixed-effects model generates the well-known Nickell bias. Still, this bias is of order (1/T) and is therefore primarily troublesome in very short panels, when T is 3 or 4 (Nickell 1981). Since the average time dimension in our data exceeds 15, this concern is mitigated. Moreover, Beck and Katz's Monte Carlo evidence shows that, for the values of (T) typically encountered in TSCS applications, roughly 20 or more, OLS/LSDV with fixed effects performs about as well as the Kiviet correction and substantially better than the Anderson-Hsiao estimator (Beck and Katz 2011). This supports the use of LSDV in our setting, especially because it remains flexible enough to accommodate country-specific intercepts and slope heterogeneity. With only 27 countries, dynamic panel GMM is not an attractive baseline estimator. Arellano-Bond and related GMM estimators are designed for large-N, small-T panels. In our setting, the advantages of GMM are limited, while the risks of instrument proliferation, weak specification tests, and loss of flexibility are substantial. We therefore prefer an LSDV-based dynamic specification.

[4] Alternative specifications with period fixed effects were also estimated and reported in Appendix A, yielding qualitatively similar results. We choose the COVID-19 dummy solution as a more parsimonious way to capture the exceptional fiscal dynamics during the pandemic period.

Given the exposure of EU countries to common macroeconomic and fiscal shocks, we apply the Pesaran CD test to the residuals of the baseline ECMs (Pesaran, 2004). Insofar as the null hypothesis of cross-sectional independence is rejected, we replace standard errors with Driscoll–Kraay standard errors, which are robust to heteroskedasticity, serial correlation, and general forms of cross-sectional dependence (Driscoll and Kraay, 1998). The covariance matrix is estimated using the Bartlett kernel proposed by Newey and West (1987), with a maximum lag order of two years. This parsimonious lag structure balances the need to accommodate short-memory serial correlation against the limited and unbalanced time dimension of the panel; the main results are also robust to maximum lag orders of one and three years. The complete diagnostic is reported in Appendix D.

Including the lagged dependent variable captures the strong persistence typically observed in public expenditure. Fixed effects control for time-invariant country characteristics such as institutional design or welfare state structure. The coefficients on changes in the explanatory variables capture contemporaneous short-run effects, while the coefficient on the lagged dependent variable in the differenced equation measures the speed at which expenditure returns to its model-implied long-run level. Although long-run multipliers are obtained as the negative ratio of the lagged-level coefficients to the adjustment coefficient α, the cluster analysis uses the country-specific lagged-level coefficients directly because the adjustment coefficient is common across countries within each expenditure model. The classification therefore focuses on cross-country heterogeneity in the contribution of ageing to the adjustment dynamics.

- α captures the speed of adjustment towards the long-run level
- β parameters capture long-run effects through lagged levels ($it-1$)
- λ parameters capture short-run effects through differenced variables (Δ)

However, a homogeneity test of slopes clearly rejects the null hypothesis of common coefficients across countries for all expenditure variables. The homogeneous specification would therefore be interpreted, at best, as an average effect across countries, potentially masking important differences in the magnitude or even the direction of the relationship. For this reason, the next step is to move to econometric approaches that allow the estimated coefficients to vary across countries. We estimate a mixed-effects (random-coefficients) model using restricted maximum likelihood (REML), which provides consistent estimates of variance components (Baltagi, 2021). This approach enables the recovery of country-specific coefficients that can subsequently be used in a clustering analysis.

The specification allows the coefficients associated with demographic variables to vary across countries[5]:

[5] We allow slope heterogeneity only for the theoretically central variables to avoid overparameterizing the model. Given the limited cross-sectional dimension of the panel, estimating random slopes for all covariates would require many additional variance-covariance parameters and could lead to unstable estimates. This restricted mixed-effects specification provides a compromise between a fully homogeneous model and a fully heterogeneous one: it allows the key effects to vary across countries while maintaining a parsimonious structure for the remaining controls.

$$\Delta \exp_{it} = \alpha \exp_{it-1} + \beta_{1i} pc65_{it-1} + \beta_{2i} p80_{it-1} + \lambda_{1i} \Delta pc65_{it} +$$

$$+ \lambda_{2i} \Delta p80_{it} + \delta' \Delta X_{it} + \delta COVID_t + \mu_i + \varepsilon_{it} \qquad [2]$$

with:

$$\beta_{ki} = \beta_k + v_{ki}, \qquad \lambda_{ki} = \lambda_k + w_{ki}$$

Third, the country-specific short- and long-run coefficients recovered from the mixed-effects model [2] are used as inputs in an agglomerative hierarchical cluster analysis. Ward's linkage method is applied separately to each expenditure category using the four country-specific coefficients associated with general and advanced ageing. The resulting dendrograms are visually inspected to identify meaningful breaks in the agglomeration structure. The coefficients are not standardized because they are expressed in comparable units and their relative magnitudes constitute part of the heterogeneity of interest.

The final step consists of re-estimating the ECM, allowing for discrete heterogeneity across clusters identified in equation [2]. This approach replaces continuous random heterogeneity with a grouped structure that improves interpretability while preserving the main patterns of cross-country differences. Hence, the ECM with clusters is now specified as:

$$\Delta \exp_{it} = \alpha \exp_{it-1} + \sum_c D_{ic} \beta_{1c} pc65_{it-1} + \sum_c D_{ic} \beta_{2c} p80_{it-1} + \sum_c D_{ic} \lambda_{1c} \Delta pc65_{it} +$$

$$+ \sum_c D_{ic} \lambda_{2c} \Delta p80_{it} + \delta' \Delta X_{it} + \delta COVID_t + \mu_i + \varepsilon_{it} \qquad [3]$$

Where:

- D_{ic} are cluster dummies (equal to 1 if country *i* belongs to cluster *c*).
- α captures the speed of adjustment towards the long-run level
- β_c are cluster-specific parameters capturing long-run effects through lagged levels $(it-1)$
- λ_c are cluster-specific parameters capturing short-run effects through differenced variables (Δ)

Countries within the same cluster share similar adjustment patterns; differences in expenditure dynamics are captured at the group level rather than the individual level; the ECM structure ensures a consistent distinction between short-run shocks and long-run equilibrium adjustment.

4. ECONOMETRIC RESULTS

Table 2 presents summary statistics for all variables. The data reveal substantial cross-country heterogeneity in both fiscal outcomes and demographic structures. Total public age-related expenditure averages around one-fifth of GDP, although the dispersion is considerable across countries and over time. Pensions account for the largest share, followed by healthcare, while long-term care remains comparatively small but is increasingly important in more advanced ageing contexts. Demographic indicators also exhibit notable variation. The share of the population aged 65 and over has increased

steadily across all countries, while the proportion of individuals aged 80 and over within the elderly population shows even greater heterogeneity, reflecting differences in longevity and population structure. Macroeconomic variables also vary significantly, particularly economic growth and fiscal balances, which are strongly influenced by business-cycle fluctuations and major economic events, including the global financial crisis and the COVID-19 pandemic. Overall, the descriptive statistics highlight both cross-sectional and temporal variation.

Moreover, **Figure 1** displays the estimated relationship between expenditure and the share of the population aged 65+ and 80+. The graphical evidence confirms heterogeneity across countries. While some show a strong positive association between ageing and expenditure, others show flatter or even weak relationships. This heterogeneity is observed in both the broader elderly population (65+) and for the oldest cohorts (80+), although the latter tends to show stronger associations in certain expenditure categories, particularly healthcare and long-term care.

<Insert Table 2>

<Insert Figure 1>

To test for unit roots in the left-hand variables, we rely upon a Fisher-type panel test based on augmented Dickey–Fuller regressions (**Table 3**). This approach suits our unbalanced TSCS panel because it allows heterogeneous dynamics across countries while remaining parsimonious in a short time dimension. In all cases except for p, the test statistics lead us to clearly reject the null of a unit root at conventional significance levels, indicating stationarity. Except for p, the test statistics clearly reject the null hypothesis of a unit root at conventional significance levels, providing evidence of stationarity. For p, the evidence is less conclusive, since the unit-root null cannot be rejected at the 10 percent level. Nevertheless, the subsequent dynamic estimations yield a negative and statistically significant coefficient on the lagged dependent variable, pointing to a corrective, mean-reverting adjustment mechanism and is consistent with dynamically stable behavior. Taken together, these findings support the use of an error-correction specification to model both short-run dynamics and adjustment toward a long-run equilibrium relationship.

<Insert Table 3>

Table 4 reports results for equations [1] and [3] across the four expenditure categories. Appendix B presents the mixed-effects estimates corresponding to equation [2]. Appendix C reports the country-specific coefficients recovered from these models and used as inputs in the clustering procedure. Inspection of the Ward hierarchical dendrograms in **Figure 2** supports a four-cluster solution for each expenditure category.

Finally, **Figure 3** displays the resulting geographical distribution of the clusters across the EU-27, and **Table 5** groups countries by cluster and expenditure category.

<Insert Table 4>

<Insert Figure 2>

<Insert Figure 3>

<Insert Table 5>

Age-Related Expenditure aggregate (a)

The baseline ECM results confirm that public expenditure shows strong inertia, as reflected in the negative and highly significant coefficient on the lagged dependent variable (−0.170). In the short run, increases in the share of the population aged 65 ($\Delta pc65_{it}$) have a positive effect, while the 80+ population has no statistically significant effect. Neither lagged ageing level is statistically significant. Overall, demographic pressure is therefore primarily associated with the contemporaneous response to general ageing, against a background of strong expenditure persistence. Looking at differences across countries, four clusters of countries emerge:

- **Cluster 1:** Austria (AT); Belgium (BE); Finland (FI); Cyprus (CY); Latvia (LV); Romania (RO); Italy (IT); Netherlands (NL); Spain (ES); Lithuania (LT); France (FR); Malta (MT); Sweden (SE). The clustered ECM results show a strong, statistically significant short-run effect of ageing (65+), indicating that increases in the elderly population rapidly translate into higher expenditure. The short-run effect of advanced ageing (80+) is not statistically significant, suggesting that the composition of ageing does not play a relevant role in immediate expenditure dynamics. In the long run, both demographic variables also display positive coefficients, although the evidence is stronger for the 65+ indicator and only marginal for the 80+ indicator. Hence, this cluster, which includes the largest number of countries, shares the baseline model's positive contemporaneous association with general ageing but differs in also displaying significant lagged-level demographic coefficients.
- **Cluster 2**: Denmark (DK) and Estonia (EE). Neither demographic variable is statistically significant in the short or long run, except for a marginal negative contemporaneous effect of advanced ageing, indicating that ageing is not consistently or stably associated with expenditure.
- **Cluster 3:** Germany (DE); Croatia (HR); Poland (PL); Portugal (PT); Slovenia (SI); Greece (EL); Ireland (IE); Luxembourg (LU). The clustered ECM reveals no short-run or long-run effects for either demographic variable.
- **Cluster 4:** Bulgaria (BG); Czechia (CZ); Slovakia (SK); Hungary (HU). The clustered ECM results reveal a negative contemporaneous association with general ageing. In the short run, a higher share of the elderly population is associated with significantly lower expenditure, indicating a negative short-run

association that may reflect fiscal restraint or other unobserved institutional responses. In the long run, neither variable is statistically significant.

Healthcare Expenditure (h)

Short-run effects of general ageing (65+) show a weak positive association with healthcare expenditure in the baseline ECM. Regarding the long-run coefficients, the 80+ population are positive and statistically significant. Macroeconomic variables behave as expected, with growth exerting a negative effect and the COVID-19 dummy capturing a strong and significant increase in spending. Four groups of countries are detected:

- **Cluster 1**: Austria (AT); Malta (MT); France (FR); Belgium (BE); Netherlands (NL); Sweden (SE); Germany (DE). The clustered ECM results reveal a clear and consistent role for advanced ageing. In the short run, the effect of the 65+ population is not statistically significant, whereas the coefficient on $\Delta p80_{it}$ is positive and highly significant. Again, in the long run, the coefficient on the 65+ population is smaller and only weakly significant, but the lagged 80+ population shows a positive and statistically significant impact.
- **Cluster 2**: Denmark (DK); Finland (FI); Ireland (IE); Spain (ES); Portugal (PT); Slovenia (SI). The clustered ECM results reveal a highly selective and asymmetric response to demographic change. In the short run, the coefficient on $\Delta pc65_{it}$ is not statistically significant, whereas the effect of $\Delta p80_{it}$ is negative and highly significant. In the long run, demographic variables show no statistically significant relationship.
- **Cluster 3:** Bulgaria (BG); Czechia (CZ); Cyprus (CY); Croatia (HR); Slovakia (SK); Italy (IT); Lithuania (LT); Latvia (LV); Poland (PL); Romania (RO). Results show a differentiated response to demographic change across time horizons. In the short run, the coefficient on $\Delta pc65_{it}$ is not statistically significant, while the effect of advanced ageing ($\Delta p80_{it}$) is positive and significant. In contrast, the long-run adjustment follows a different pattern. The coefficient on the lagged 65+ population is positive and statistically significant, whereas the coefficient on the 80+ variable is not.
- **Cluster 4:** Estonia (EE); Hungary (HU); Greece (EL); Luxembourg (LU). The clustered ECM results reveal a clear but asymmetric response to demographic change. In the long run, the coefficient of advanced ageing is positive and statistically significant. By contrast, no significant effect is observed for the broader 65+ population in either the short or long run.

The negative and statistically significant contemporaneous coefficient on advanced ageing in Cluster 2 and Cluster 4 indicate that demographic change is not uniformly associated with immediate expenditure growth. These patterns may be consistent with different mechanisms, including expenditure constraints, efficiency improvements, delayed institutional responses, or differences in service provision. For Cluster 2, the negative short-run effect on the 80+ population could suggests that countries with more developed public healthcare systems have greater capacity to implement efficiency measures as demographic pressure increases, preventing public expenditure from rising in proportion. In Cluster 4, the negative-to-positive sign could reflect a delayed adjustment: short-run fiscal constraints temporarily prevent the expenditure response to demographic pressure, but long-run equilibrium converges to the expected positive relationship as institutional adjustments overcome financing constraints.

Long-Term Care Expenditure (ltc)

In the baseline ECM, demographic effects are not statistically significant in either the short or long run. The results change slightly when the analysis is broken down by country group.

- **Cluster 1**: Austria (AT); Romania (RO); Belgium (BE); Hungary (HU); Bulgaria (BG); Slovenia (SI); Denmark (DK); Croatia (HR); Portugal (PT); Cyprus (CY); Czechia (CZ); Poland (PL); Italy (IT); Latvia (LV); Sweden (SE). In the short run, the coefficient on $\Delta p80_{it}$ is negative and statistically significant, indicating that increases in the oldest population are associated with immediate expenditure restraint. In the long run, neither demographic variable shows a statistically significant effect, suggesting no stable structural link between ageing and LTC expenditure.
- **Cluster 2**: Estonia (EE); Ireland (IE); Greece (EL). In the short run, the coefficient on $\Delta p80_{it}$ is again negative and highly significant, but much larger than in the case of Cluster 1. Among the lagged ageing levels, the 65+ coefficient is negative and statistically significant, while the 80+ coefficient is positive but only marginally significant. This cluster therefore combines a negative immediate response to advanced ageing with differentiated contributions from the lagged demographic levels.
- **Cluster 3:** Luxembourg (LU); Malta (MT). Both contemporaneous demographic coefficients are negative, but only the coefficient associated with advanced ageing is marginally significant. Neither lagged-level demographic coefficient is statistically significant.
- **Cluster 4:** Germany (DE); Finland (FI); France (FR); Spain (ES); Netherlands (NL); Lithuania (LT); Slovakia (SK). In the short run, the coefficient on $\Delta p80_{it}$ is positive and significant. Among the lagged demographic levels, the 80+ coefficient is also positive and statistically significant, while the 65+ coefficient is not. Advanced ageing is therefore positively associated with both the immediate response and the adjustment dynamics of LTC expenditure in this cluster.

Pension Expenditure (p)

Results for the baseline ECM indicate a positive and highly significant contemporaneous association between the population aged 65 and over and pension expenditure. By contrast, the contemporaneous coefficient associated with advanced ageing is not statistically significant. Among the lagged demographic levels, the coefficient associated with the 65+ population is negative and statistically significant, whereas the coefficient associated with the 80+ indicator is not. Results for the clusters are as follows:

- **Cluster 1:** Austria (AT); Netherlands (NL); Croatia (HR); Sweden (SE); Spain (ES); Germany (DE); Slovakia (SK); Luxembourg (LU); Malta (MT); Slovenia (SI). In the short run, the coefficient on $\Delta pc65_{it}$ is positive and statistically significant. In contrast, the effect of advanced ageing ($\Delta p80_{it}$) is negative and significant. In the long run, the dominant result is a negative and highly significant coefficient on the lagged 65+ population. This negative association may be consistent with pension adjustment mechanisms, whereby pension systems adjust

over time to offset the fiscal impact of population ageing. Positive but only marginally significant long-run effect is observed for the 80+ population.

- **Cluster 2:** Belgium (BE); Portugal (PT); Poland (PL); Cyprus (CY); Estonia (EE); France (FR); Italy (IT); Greece (EL); Romania (RO). Both $\Delta pc65_{it}$ and $\Delta p80_{it}$ exhibit positive and statistically significant coefficients, indicating that both contemporaneous demographic coefficients are positive and statistically significant, indicating that increases in general and advanced ageing are associated with and immediate increase in pension expenditure as a percentage of GDP. This suggests strong short-term demographic responsiveness. Among the lagged ageing levels, the coefficient associated with the 65+ population is negative and statistically significant, whereas the coefficient associated with the 80+ indicator is not. The cluster therefore combines strong positive contemporaneous responsiveness with a negative lagged-level contribution from general ageing.
- **Cluster 3:** Denmark (DK); Finland (FI); Lithuania (LT); Latvia (LV). The coefficient on $\Delta pc65_{it}$ is large and highly significant, indicating that increases in the elderly population are rapidly and substantially reflected in pension expenditure. By contrast, the contemporaneous coefficient associated with advanced ageing is negative but not statistically significant. Neither demographic variable shows significant long-run effects, indicating no structural adjustment mechanism linking ageing to expenditure over time. The demographic association identified for this cluster is therefore concentrated in the contemporaneous response to general ageing.
- **Cluster 4:** Bulgaria (BG); Czechia (CZ); Ireland (IE); Hungary (HU). The clustered ECM results reveal that none of the contemporaneous or lagged-level ageing coefficients are statistically significant.

The joint results for healthcare, long-term care (LTC), and pension expenditure reveal three main patterns. First, demographic effects are markedly heterogeneous across countries and spending categories. Contemporaneous responses are particularly evident in pensions, whereas healthcare and LTC respond more selectively, especially to advanced ageing. Second, advanced ageing (80+) plays a prominent but uneven role, primarily in healthcare and LTC dynamics, appearing through both contemporaneous responses and lagged-level coefficients depending on the country cluster. In pensions, general ageing is more consistently associated with expenditure, while the contribution of advanced ageing is more fragmented across country groups.

Third, the cluster analysis points to a certain geographical and institutional structure: Central and Eastern European economies tend to exhibit weaker or more delayed responses,[6] whereas Western and Nordic countries tend to show stronger and more persistent adjustments.[7] This pattern is more evident in healthcare and LTC than in pensions, where expenditure shows greater persistence and heterogeneity in adjustment

[6] Bulgaria (BG), Czechia (CZ), Hungary (HU), Slovakia (SK), and, in an outer, less cohesive circle: Latvia (LV), Lithuania (LT), Poland (PL), Croatia (HR), and Romania (RO).
[7] Austria (AT), Belgium (BE), Denmark (DK), Finland (FI), and, in a second, more permissive circle, France (FR), Germany (DE), the Netherlands (NL), and Sweden (SE).

patterns, reflecting the decisive role of institutional design in shaping public spending responses to ageing.

5. CONCLUSIONS

A central finding of this paper is that the relationship between population ageing and age-related public expenditure is highly heterogeneous across expenditure categories, demographic dimensions, time horizons, and countries. Rather than following a uniform pattern, the results reveal a set of distinct and sometimes opposing adjustment patterns, suggesting that differences in institutional design, policy frameworks, and fiscal capacity may shape the fiscal implications of ageing.

Another robust finding across the empirical specifications is the strong persistence of public expenditure. This persistence is especially pronounced in pensions and healthcare, where legal commitments and structural entitlements shape expenditure. By contrast, long-term care shows weaker, less stable dynamics, consistent with a more heterogeneous, evolving institutional structure.

Third, the 80+ population, which captures advanced ageing and care intensity, is a key driver of healthcare, but its impact on long-term care and pensions is weaker and less systematic.

Fourth, the clustering analysis shows several differentiated expenditure-response patterns, ranging from expansionary and reactive responses to more constrained or weakly responsive trajectories. These results suggest that demographic pressures are filtered through heterogeneous institutional and fiscal contexts rather than producing uniform expenditure outcomes. Despite this heterogeneity, some regularities emerge. Several Western and Northern European countries consistently cluster together across healthcare, long-term care, and pension expenditure, reflecting broadly similar welfare-state structures and stronger institutional capacity to respond to demographic pressures. In contrast, many Central and Eastern European countries tend to form a distinct group characterized by weaker or more delayed expenditure responses, consistent with more limited fiscal capacity and less consolidated welfare arrangements. However, this structural stability is only partial. Cluster membership varies substantially across expenditure categories, particularly in long-term care and pensions, where institutional designs differ more markedly across countries. Smaller states, transition economies, and countries with hybrid welfare configurations show the highest cluster mobility, often shifting groups depending on the expenditure category.

While this study offers valuable insights, several empirical limitations warrant acknowledgement. As in the broader literature, identifying the underlying mechanisms poses significant econometric challenges, and identification remains a concern. Consequently, researchers should not automatically interpret the estimated coefficients as definitive causal effects. Future research should use microdata and more sophisticated identification strategies, including regression discontinuity designs and other quasi-experimental approaches, to uncover the mechanisms linking ageing to public expenditure outcomes. Such approaches would be particularly valuable for assessing whether commonly used age thresholds, such as 65 and 80 years, capture meaningful behavioural and fiscal discontinuities or whether their relevance varies systematically across countries, institutional settings, and expenditure functions.

Finally, these findings have differentiated implications for public policy. For pension systems, the strong persistence of expenditure and its greater short-run sensitivity to general ageing highlight the importance of considering demographic pressures alongside existing benefit commitments and pension adjustment mechanisms. The diversity of policy responses provides opportunities for countries to learn from those that have managed demographic pressures more effectively. In healthcare, the stronger association with advanced ageing, particularly in the long run, suggests that expenditure planning should account for changes in the age composition of the elderly population. This calls for greater organisational flexibility and stronger emphasis on efficiency. For long-term care, the lack of a systematic relationship between demographic ageing and expenditure in the baseline estimates, along with substantial cross-country heterogeneity, underscores the importance of differences in coverage, service provision, and financing arrangements. The relatively limited development of long-term care systems in some countries offers opportunities to incorporate more sustainable arrangements as these systems expand. More broadly, policy responses should reflect national demographic and institutional circumstances rather than follow uniform approaches. In this respect, the identified clusters should be interpreted as expenditure-specific adjustment patterns rather than fixed institutional categories, as country membership varies across expenditure functions.

Data Availability Statement

The data and Stata code required to replicate the results reported in this paper are publicly available from the Harvard Dataverse repository at https://doi.org/10.7910/DVN/LAQZZP

Declaration of generative AI use

The authors used and AI tool for copyediting and language refinement, reference management, identifying and organising potentially relevant literature, and software coding assistance. All AI-assisted outputs, references, and code were reviewed and verified by the authors. The AI tool was not used to generate the study's arguments, interpret the results, or draw its conclusions. They take full responsibility for the originality, accuracy, and integrity of the manuscript.

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Table 1. Variable definitions

Variable	Description	Measurement	Source	Period
a	Total public age-related expenditure [h + ltc + p]	% of GDP	Eurostat	1992-2023
h	Public expenditure on Healthcare	% of GDP	Eurostat	1992-2024
ltc	Public expenditure on Long-Term Care	% of GDP	Eurostat	1992-2024
p	Public expenditure on Pensions	% of GDP	Eurostat	1992-2023
pc65	Old-age dependency ratio: ratio between the population aged 65+ and those aged 15-64	% of total population	Eurostat	1992-2024
p80	Age intensity ratio: ratio between the population aged 80+ and those aged 65+	% of population aged 65 and over	Eurostat	1992-2024
growth	Real GDP growth rate	percentage point change	Eurostat	1992-2024
gdp	GDP per capita in PPS	EU27 = 100	Eurostat	1995-2024
exp_gdp	General government expenditure	% of GDP	Eurostat	1992-2024
def	Net lending / net borrowing	% of GDP	Eurostat	1995-2024

Note: The aggregate measure is calculated as a=h+ltc+p only for country-year observations in which all three components are available.

Table 2. Descriptive statistics

	N	Average-T	Mean	Std. Dev.	Min	Max
Total public age-related expenditure (% of GDP)	451	17	20.58	4.70	11.19	30.14
Public expenditure on Healthcare (% of GDP)	461	17	8.59	1.81	4.47	12.70
Public expenditure on Long-Term Care (% of GDP)	456	17	1.13	0.83	0.01	3.20
Public expenditure on Pensions (% of GDP)	744	28	10.37	2.80	3.66	18.17
Share of population aged 65 and over	882	33	16.56	3.09	10.30	24.30
Share of population aged 80 and over relative to population aged 65+	881	33	23.75	3.93	12.98	33.33
Real GDP growth rate (percentage point change)	793	29	2.56	3.73	-16.04	24.62
GDP per capita in PPS	810	30	24,884	13,638	4,477	97,689
General government expenditure (% of GDP)	798	30	45.47	7.05	20.70	64.90
Net lending / net borrowing (% of GDP)	808	30	-2.65	3.49	-32.10	6.90

Table 3. Unit Root test results

ADF-Fisher	**a**	**h**	**ltc**	**p**
p-value	0.0001	0.0000	0.0217	0.1067
Statistic	3.7107	5.5875	2.0200	1.2444

Notes: Fisher-type panel unit root tests based on ADF regressions with 1 lag. Null hypothesis: all panels contain a unit root. Modified inverse Chi-squared statistics and p-values are reported. Panels are unbalanced.

Table 4. Estimation results

	age-related expenditure [a]		Healthcare expenditure [h]		Long-term-care expenditure [ltc]		Pension expenditure [p]	
	ECM [1]	**Clustered ECM [3]**	**ECM [1]**	**Clustered ECM [3]**	**ECM [1]**	**Clustered ECM [3]**	**ECM [1]**	**Clustered ECM [3]**
Lag	-0.170***	-0.225***	-0.253***	-0.297***	-0.133***	-0.159***	-0.124***	-0.139***
	(-5.91)	(-7.79)	(-4.11)	(-3.93)	(-4.80)	(-4.81)	(-5.44)	(-5.40)
Δpc65	0.404***		0.118*		-0.018		0.236***	
	(3.31)		(1.97)		(-0.65)		(3.11)	
Δp80	0.040		0.034		-0.005		0.023	
	(1.11)		(1.03)		(-0.80)		(1.04)	
L.pc65	0.018		0.020		0.003		-0.026**	
	(0.86)		(1.45)		(0.71)		(-2.23)	
L.p80	0.024		0.020**		0.003		0.000	
	(1.07)		(2.33)		(1.54)		(0.04)	
growth	-0.172***	-0.171***	-0.054***	-0.054***	-0.009***	-0.009***	-0.108***	-0.109***
	(-31.32)	(-23.00)	(-9.08)	(-9.36)	(-7.93)	(-9.43)	(-25.29)	(-25.37)
L.gdp	-0.000**	-0.000***	-0.000	-0.000	0.000	0.000	-0.000	-0.000
	(-2.74)	(-3.03)	(-0.97)	(-1.38)	(1.16)	(1.51)	(-0.85)	(-1.51)
L.exp_gdp	-0.007	0.014	0.010	0.006	0.001	0.001	0.004	0.001
	(-0.37)	(0.81)	(0.82)	(0.51)	(0.46)	(0.57)	(0.45)	(0.07)
L.def	0.013	0.032**	0.021**	0.011	0.004*	0.004	0.007	0.001
	(1.12)	(2.59)	(2.37)	(1.24)	(1.74)	(1.68)	(0.84)	(0.18)
covid	0.692***	0.727***	0.560***	0.543***	0.016***	0.014**	0.115***	0.123***
	(8.23)	(6.92)	(7.97)	(7.60)	(3.18)	(2.51)	(3.62)	(4.20)
Δpc65 by clusters								
1		0.775***		0.162		-0.003		0.131*
		(4.34)		(1.43)		(-0.11)		(1.80)
2		0.701		0.022		0.043		0.485***
		(1.31)		(0.16)		(1.03)		(8.28)
3		0.076		0.155		-0.078		0.628**
		(0.31)		(1.15)		(-0.31)		(2.48)
4		-1.090***		0.058		-0.010		-0.487
		(-3.86)		(0.37)		(-0.20)		(-1.61)
Δp80 by clusters								
1		0.061		0.077**		-0.011***		-0.049**
		(1.36)		(2.60)		(-2.91)		(-2.11)
2		-0.165*		-0.116***		-0.045***		0.115***
		(-1.79)		(-2.91)		(-3.79)		(3.16)
3		0.112		0.154***		-0.113*		-0.137
		(0.97)		(4.58)		(-1.80)		(-1.51)

	age-related expenditure [a]		Healthcare expenditure [h]		Long-term-care expenditure [ltc]		Pension expenditure [p]	
	ECM [1]	Clustered ECM [3]	ECM [1]	Clustered ECM [3]	ECM [1]	Clustered ECM [3]	ECM [1]	Clustered ECM [3]
4		0.223		-0.071		0.028**		0.088
		(0.83)		(-1.01)		(2.54)		(1.30)
L.pc65 by clusters								
1		0.064**		0.038*		0.004		-0.031***
		(2.31)		(1.85)		(0.97)		(-4.92)
2		-0.067		0.009		-0.020**		-0.018**
		(-0.39)		(0.49)		(-2.17)		(-2.09)
3		0.005		0.070***		-0.013		-0.015
		(0.18)		(4.31)		(-0.21)		(-0.62)
4		-0.001		-0.023		0.005		0.000
		(-0.02)		(-0.80)		(0.73)		(0.02)
L.p80 by clusters								
1		0.065*		0.046***		0.002		0.014*
		(1.82)		(4.66)		(0.88)		(1.85)
2		0.038		0.015		0.007*		0.005
		(0.63)		(1.17)		(1.92)		(0.72)
3		0.035		0.016		-0.040		-0.005
		(1.24)		(0.75)		(-0.78)		(-0.27)
4		-0.022		0.031**		0.007**		-0.005
		(-0.19)		(2.59)		(2.21)		(-0.23)
_cons	3.639***	3.347***	1.081**	1.266***	-0.014	0.056	1.836***	1.914***
	(4.76)	(3.04)	(2.47)	(2.84)	(-0.15)	(0.34)	(6.15)	(6.37)
N	421	421	431	431	426	426	686	686
Countries	27	27	27	27	27	27	27	27
Average T	15.59	15.59	15.96	15.96	15.78	15.78	25.41	25.41
Max VIF	3.49	3.49	2.21	2.21	1.91	1.91	2.77	2.77
R^2 within	0.75	0.77	0.64	0.66	0.30	0.36	0.64	0.67
R^2 between	0.04	0.13	0.05	0.17	0.02	0.00	0.00	0.01
R^2 overall	0.36	0.23	0.27	0.16	0.02	0.01	0.47	0.49

Notes: Driscoll–Kraay t-statistics, robust to heteroskedasticity, serial correlation, and cross-sectional dependence, are reported in parentheses. The variance–covariance matrix uses a Bartlett kernel with a maximum lag order of two years. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All specifications include country fixed effects. The maximum VIF excludes the lagged dependent variable..

Table 5. Country clusters by expenditure category

	Aggregate (A)	Healthcare (H)	Long-Term-Care (LTC)	Pensions (P)
Cluster 1	Austria (AT), Belgium (BE), Cyprus (CY), Spain (ES), Finland (FI), France (FR), Italy (IT), Lithuania (LT), Latvia (LV), Malta (MT), Netherlands (NL), Romania (RO), Sweden (SE)	Austria (AT), Belgium (BE), Germany (DE), France (FR), Malta (MT), Netherlands (NL), Sweden (SE)	Austria (AT), Belgium (BE), Bulgaria (BG), Cyprus (CY), Czechia (CZ), Denmark (DK), Croatia (HR), Hungary (HU), Italy (IT), Latvia (LV), Poland (PL), Portugal (PT), Romania (RO), Sweden (SE), Slovenia (SI)	Austria (AT), Germany (DE), Spain (ES), Croatia (HR), Luxembourg (LU), Malta (MT), Netherlands (NL), Sweden (SE), Slovenia (SI), Slovakia (SK)
Cluster 2	Denmark (DK), Estonia (EE)	Denmark (DK), Spain (ES), Finland (FI), Ireland (IE), Portugal (PT), Slovenia (SI)	Estonia (EE), Greece (EL), Ireland (IE)	Belgium (BE), Cyprus (CY), Estonia (EE), Greece (EL), France (FR), Italy (IT), Poland (PL), Portugal (PT), Romania (RO)
Cluster 3	Germany (DE), Greece (EL), Croatia (HR), Ireland (IE), Luxembourg (LU), Poland (PL), Portugal (PT), Slovenia (SI)	Bulgaria (BG), Cyprus (CY), Czechia (CZ), Croatia (HR), Italy (IT), Lithuania (LT), Latvia (LV), Poland (PL), Romania (RO), Slovakia (SK)	Luxembourg (LU), Malta (MT)	Denmark (DK), Finland (FI), Lithuania (LT), Latvia (LV)
Cluster 4	Bulgaria (BG), Czechia (CZ), Hungary (HU), Slovakia (SK)	Estonia (EE), Greece (EL), Hungary (HU), Luxembourg (LU)	Germany (DE), Spain (ES), Finland (FI), France (FR), Lithuania (LT), Netherlands (NL), Slovakia (SK)	Bulgaria (BG), Czechia (CZ), Hungary (HU), Ireland (IE)

Note: Clusters are derived from hierarchical clustering (Ward's method) applied to country-specific coefficients extracted from the mixed-effects ECM model.

Figure 1. Slope heterogeneity

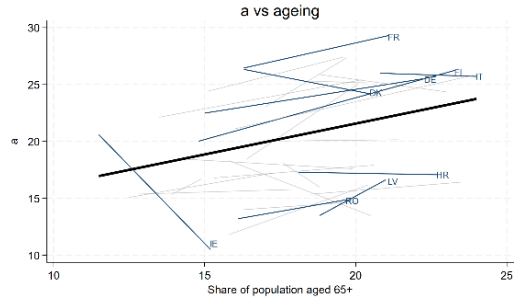
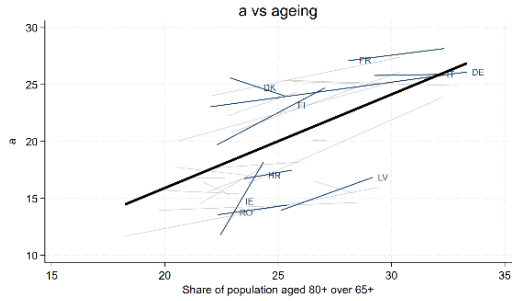
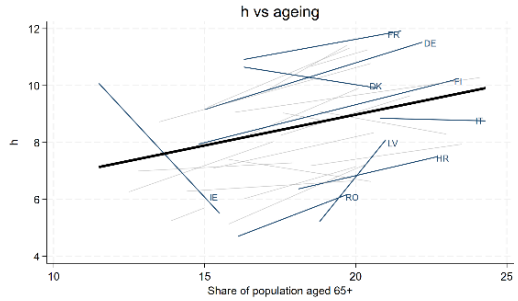
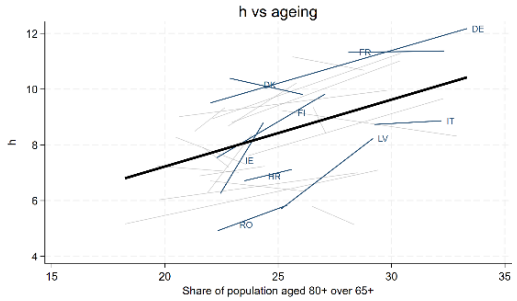
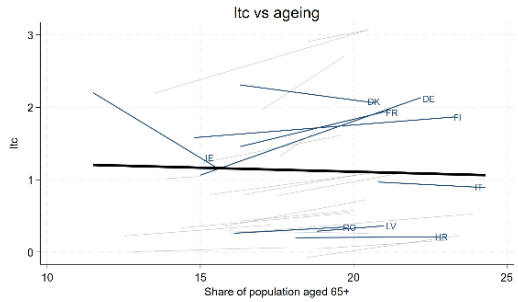
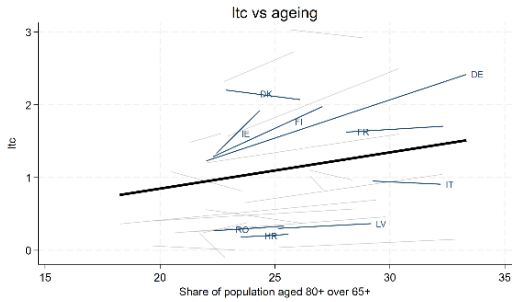
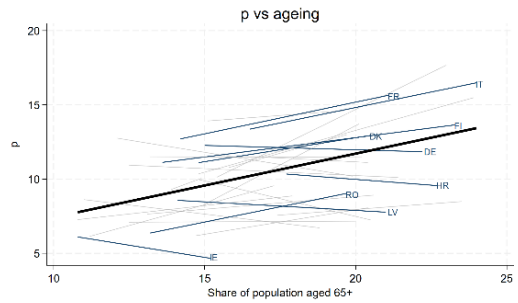
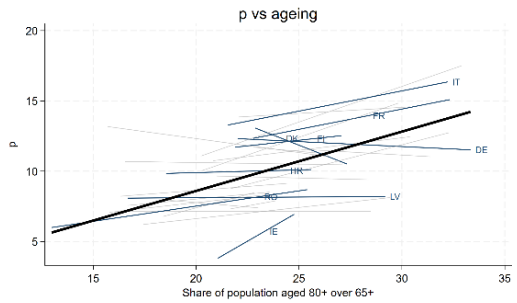
Total public age-related expenditure (% of GDP)	
Share of population aged 65 and over	*Share of population aged 80 and over relative to population aged 65+*
	
Public expenditure on Healthcare (% of GDP)	
Share of population aged 65 and over	*Share of population aged 80 and over relative to population aged 65+*
	
Public expenditure on Long-Term Care (% of GDP)	
Share of population aged 65 and over	*Share of population aged 80 and over relative to population aged 65+*
	
Public expenditure on Pensions (% of GDP)	
Share of population aged 65 and over	*Share of population aged 80 and over relative to population aged 65+*
	

Figure 2. Dendrogram based on mixed-effects ECM coefficients

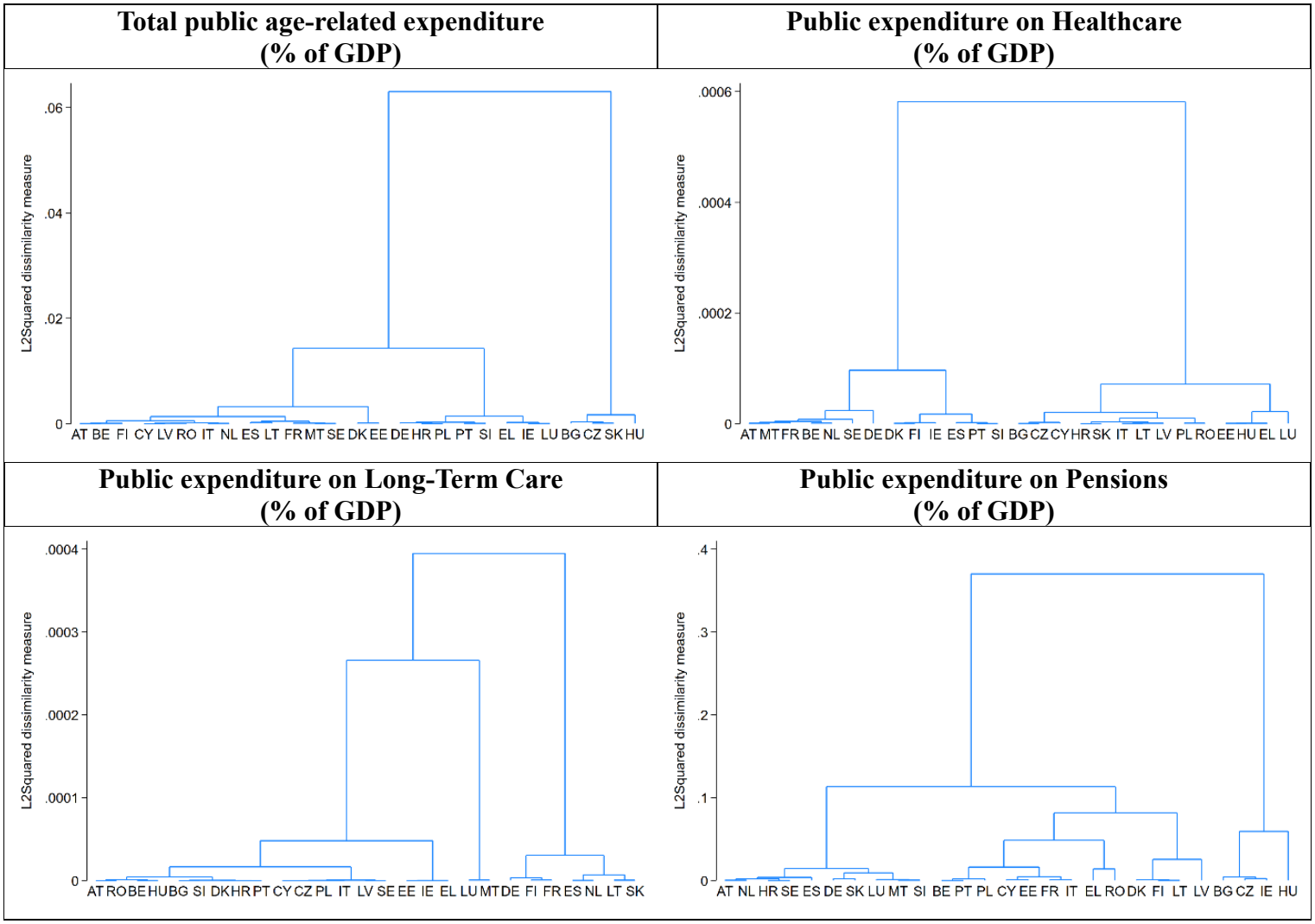
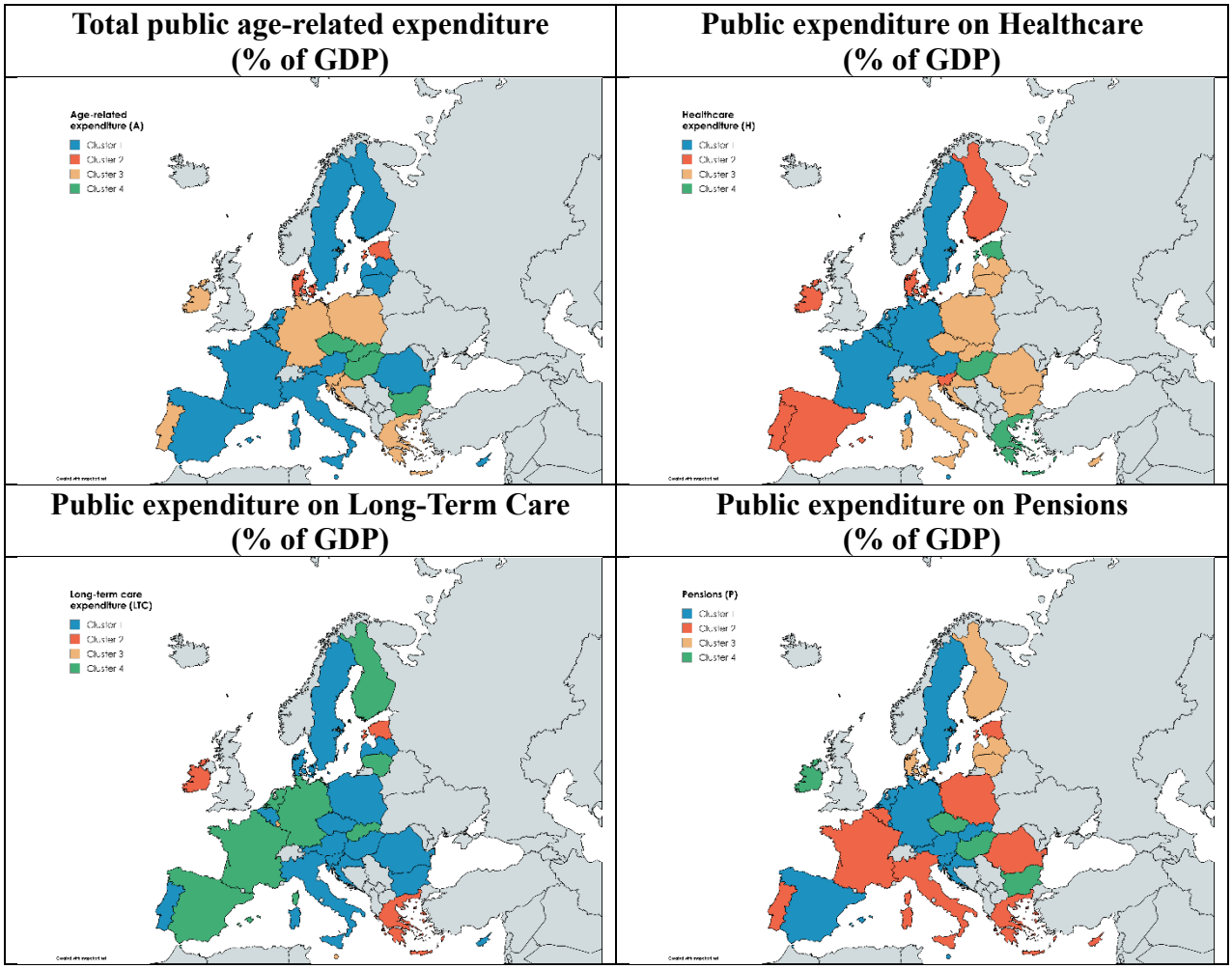


Figure 3. Maps of clustered countries



Appendix A. Fixed-Effects estimates

This appendix assesses the robustness of the baseline ECM results to the inclusion of year fixed effects. These effects absorb annual shocks common to all EU countries, including macroeconomic fluctuations and policy developments. The estimates remain qualitatively consistent with the baseline results, supporting the robustness of the main conclusions to alternative controls for common time-specific factors.

	a	**h**	**ltc**	**p**
Lag	-0.173***	-0.230***	-0.139***	-0.127***
	(-6.78)	(-4.56)	(-4.77)	(-5.51)
Δpc65	0.352**	0.072	-0.020	0.189***
	(2.71)	(0.94)	(-0.79)	(3.00)
Δp80	0.044	-0.000	-0.007	0.022
	(0.99)	(-0.01)	(-0.96)	(0.89)
L.pc65	-0.007	0.010	0.003	-0.053***
	(-0.15)	(0.40)	(0.37)	(-2.84)
L.p80	0.006	0.012	0.003	-0.003
	(0.21)	(0.88)	(1.01)	(-0.41)
growth	-0.154***	-0.058***	-0.006**	-0.094***
	(-13.94)	(-7.35)	(-2.08)	(-13.82)
L.gdp	-0.000**	-0.000*	0.000	-0.000***
	(-2.07)	(-1.88)	(1.30)	(-2.95)
L.exp_gdp	-0.004	0.009	0.002	-0.001
	(-0.21)	(0.76)	(0.81)	(-0.12)
L.def	-0.004	0.027**	0.003	-0.009
	(-0.29)	(2.16)	(1.16)	(-0.78)
Year Fixed Effect				
1997	-0.283***	-0.205***	-0.026***	0.032
	(-7.38)	(-7.25)	(-3.75)	(1.27)
1998	0.030	-0.022	0.000	0.062*
	(0.48)	(-0.53)	(0.00)	(1.77)
1999	0.024	-0.047	-0.013	0.136***
	(0.29)	(-1.10)	(-1.08)	(3.70)
2000	-0.191	-0.165***	-0.040**	0.059*
	(-1.60)	(-3.10)	(-2.05)	(2.00)
2001	-0.313**	-0.101	-0.007	0.049
	(-2.21)	(-1.31)	(-0.30)	(1.70)
2002	-0.088	0.054	0.037*	0.140***
	(-0.63)	(0.70)	(1.79)	(5.88)
2003	-0.008	0.080	0.018	0.103***
	(-0.06)	(1.11)	(1.07)	(3.88)
2004	-0.151	0.092	-0.002	0.131***
	(-0.92)	(1.15)	(-0.07)	(3.82)
2005	-0.279	0.034	0.002	0.169***
	(-1.64)	(0.38)	(0.08)	(4.84)
2006	-0.202	-0.056	0.051**	0.183***
	(-1.32)	(-0.60)	(2.36)	(4.43)
2007	-0.123	0.003	0.013	0.229***
	(-0.66)	(0.03)	(0.49)	(4.89)
2008	0.231	0.162	0.035	0.349***
	(1.19)	(1.38)	(1.46)	(7.21)
2009	0.483**	0.272*	0.079***	0.524***

	a	h	ltc	p
	(2.16)	(1.88)	(3.38)	(7.64)
2010	0.015	0.209*	0.023	0.241**
	(0.07)	(1.86)	(1.00)	(2.50)
2011	-0.024	0.073	0.007	0.158
	(-0.11)	(0.57)	(0.29)	(1.49)
2012	-0.058	0.104	0.037*	0.190*
	(-0.26)	(0.75)	(2.04)	(2.04)
2013	0.079	0.081	0.016	0.346***
	(0.33)	(0.56)	(0.80)	(3.59)
2014	0.124	0.094	0.020	0.343***
	(0.49)	(0.63)	(0.87)	(3.28)
2015	0.096	0.115	0.000	0.327***
	(0.36)	(0.75)	(0.01)	(2.90)
2016	0.150	0.167	0.019	0.312**
	(0.53)	(1.01)	(0.67)	(2.77)
2017	0.114	0.093	0.021	0.378***
	(0.38)	(0.52)	(0.64)	(3.15)
2018	0.201	0.142	0.015	0.425***
	(0.64)	(0.76)	(0.43)	(3.51)
2019	0.314	0.222	0.011	0.473***
	(0.94)	(1.14)	(0.30)	(3.79)
2020	0.908**	0.549**	0.052*	0.688***
	(2.37)	(2.39)	(1.72)	(5.78)
2021	0.721*	0.834***	0.015	0.388*
	(1.98)	(4.59)	(0.28)	(1.97)
2022	-0.363	-0.059	-0.013	0.229
	(-0.93)	(-0.28)	(-0.28)	(1.32)
2023	0.283	0.090	0.039	0.637***
	(0.69)	(0.41)	(0.91)	(3.85)
2024		0.385	0.015	
		(1.68)	(0.30)	
_cons	4.179***	1.315	-0.077	2.508***
	(2.87)	(1.67)	(-0.44)	(6.27)
N	421	431	426	686
Countries	27	27	27	27
Average T	15.59	15.96	15.78	25.41
R^2 within	0.79	0.70	0.35	0.67
R^2 between	0.01	0.06	0.02	0.00
R^2 overall	0.36	0.33	0.02	0.39

Note: Driscoll–Kraay t-statistics, robust to heteroskedasticity, serial correlation, and cross-sectional dependence, are reported in parentheses. The variance–covariance matrix uses a Bartlett kernel with a maximum lag order of two years.. All specifications include country and year fixed effects. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Appendix B. Mixed-effects estimates

This appendix reports the mixed-effects estimates underlying the cluster analysis. The models allow the short- and long-run coefficients associated with both ageing indicators to vary across countries. The reported estimations provide the basis for assessing the extent of cross-country heterogeneity and recovering the country-specific coefficients used in the clustering procedure.

	a [2]		**h** [2]		**ltc** [2]		**p** [2]	
Lag	-0.047		-0.147		-0.002		-0.114	
	(-2.86)	***	(-6.51)	***	(-0.33)		(-8.46)	***
Δpc65	0.125		0.093		-0.039		0.177	
	(0.81)		(1.08)		(-1.88)	*	(1.98)	**
Δp80	0.041		0.026		-0.010		0.011	
	(0.96)		(1.07)		(-1.56)		(0.37)	
L.pc65	-0.046		-0.004		0.001		-0.023	
	(-3.19)	***	(-0.45)		(0.34)		(-2.53)	**
L.p80	-0.000		0.009		0.000		-0.001	
	(-0.03)		(1.23)		(0.04)		(-0.25)	
growth	-0.173		-0.055		-0.009		-0.109	
	(-28.20)	***	(-15.58)	***	(-9.59)	***	(-32.02)	***
L.gdp	-0.000		-0.000		-0.000		-0.000	
	(-1.83)	*	(-1.02)		(-1.37)		(-0.45)	
L.exp_gdp	-0.003		0.010		-0.001		0.008	
	(-0.25)		(1.49)		(-1.10)		(1.33)	
L.def	0.045		0.034		0.004		0.010	
	(4.35)	***	(5.00)	***	(3.00)	***	(1.64)	
covid	0.696		0.564		0.015		0.102	
	(9.83)	***	(14.24)	***	(1.49)		(2.16)	**
_cons	2.478		0.847		0.101		1.501	
	(5.94)	***	(3.17)	***	(2.31)	**	(6.85)	***
	Estimate	***Std. err.***	***Estimate***	***Std. err.***	***Estimate***	***Std. err.***	***Estimate***	***Std. err.***
Var(Δpc65)	0.025	0.104	0.000	0.000	0.000	0.000	0.042	0.046
Var(Δpc80)	0.000	0.000	0.0002	0.003	0.0001	0.0003	0.010	0.006
Var(L.pc65)	0.000	0.000	0.000	0.000	0.000	0.000	$5.59e^{-13}$	$8.66e^{-10}$
Var(L.p80)	0.0001	0.00001	0.00003	0.00006	8.03e-^{8}	$1.06e^{-6}$	0.000	0.00003
Var(cons)	0.000	0.000	0.026	0.044	0.0002	0.0007	0.016	0.017
Var(ε)	0.204	0.017	0.064	0.005	0.004	0.0003	0.090	0.005
N	421		431		426		686	
Countries	27		27		27		27	
Average T	15.59		15.96		15.78		25.41	
Wald χ^2	1047.23		631.29		123.51		1186.56	
Wald p-value	<0.001		<0.001		<0.001		<0.001	

Note: z-statistics are reported in parentheses. Random-slope variance components and their standard errors are reported in the lower panel. Models are estimated by restricted maximum likelihood (REML), assuming an independent covariance structure for the random effects. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Appendix C. Country-specific coefficients underlying clustering

Country-specific coefficients recovered from the mixed-effects model. Short-run coefficients correspond to first-difference ageing variables (Δ65+ and Δ80+), while long-run coefficients correspond to lagged ageing variables (L.65+ and L.80+). These coefficients constitute the inputs used in the hierarchical clustering procedure. Identical coefficients across countries indicate negligible estimated cross-country variation in the corresponding random slope.

Total public age-related expenditure (a)					
Country	**Short-run 65+**	**Short-run 80+**	**Long-run 65+**	**Long-run 80+**	**Cluster**
AT	0.141	0.041	-0.046	0.009	1
BE	0.137	0.041	-0.046	0.011	1
BG	0.025	0.041	-0.046	-0.001	4
CY	0.147	0.041	-0.046	-0.004	1
CZ	0.037	0.041	-0.046	-0.005	4
DE	0.115	0.041	-0.046	0.006	3
DK	0.172	0.041	-0.046	-0.005	2
EE	0.182	0.041	-0.046	-0.007	2
EL	0.094	0.041	-0.046	-0.005	3
ES	0.167	0.041	-0.046	0.008	1
FI	0.135	0.041	-0.046	0.005	1
FR	0.149	0.041	-0.046	0.013	1
HR	0.119	0.041	-0.046	-0.005	3
HU	0.067	0.041	-0.046	-0.017	4
IE	0.110	0.041	-0.046	-0.005	3
IT	0.142	0.041	-0.046	0.004	1
LT	0.159	0.041	-0.046	-0.005	1
LU	0.105	0.041	-0.046	-0.006	3
LV	0.149	0.041	-0.046	-0.004	1
MT	0.151	0.041	-0.046	0.008	1
NL	0.148	0.041	-0.046	0.003	1
PL	0.119	0.041	-0.046	-0.002	3
PT	0.126	0.041	-0.046	0.005	3
RO	0.140	0.041	-0.046	-0.004	1
SE	0.156	0.041	-0.046	0.005	1
SI	0.129	0.041	-0.046	-0.000	3
SK	0.043	0.041	-0.046	-0.012	4
Public expenditure on Healthcare (h)					
Country	**Short-run65+**	**Short-run80+**	**Long-run65+**	**Long-run80+**	**Cluster**
AT	0.093	0.025	-0.004	0.014	1
BE	0.093	0.027	-0.004	0.014	1
BG	0.093	0.027	-0.004	0.008	3
CY	0.093	0.028	-0.004	0.009	3
CZ	0.093	0.026	-0.004	0.008	3
DE	0.093	0.030	-0.004	0.014	1
DK	0.093	0.024	-0.004	0.009	2
EE	0.093	0.024	-0.004	0.005	4
EL	0.093	0.025	-0.004	0.005	4

ES	0.093	0.026	-0.004	0.012	2
FI	0.093	0.025	-0.004	0.010	2
FR	0.093	0.026	-0.004	0.015	1
HR	0.093	0.026	-0.004	0.006	3
HU	0.093	0.024	-0.004	0.005	4
IE	0.093	0.024	-0.004	0.010	2
IT	0.093	0.026	-0.004	0.007	3
LT	0.093	0.026	-0.004	0.007	3
LU	0.093	0.024	-0.004	0.001	4
LV	0.093	0.026	-0.004	0.007	3
MT	0.093	0.026	-0.004	0.015	1
NL	0.093	0.026	-0.004	0.014	1
PL	0.093	0.027	-0.004	0.006	3
PT	0.093	0.026	-0.004	0.011	2
RO	0.093	0.028	-0.004	0.006	3
SE	0.093	0.024	-0.004	0.015	1
SI	0.093	0.025	-0.004	0.011	2
SK	0.093	0.025	-0.004	0.006	3
Public expenditure on Long-Term Care (ltc)					
Country	**Short-run65+**	**Short-run80+**	**Long-run65+**	**Long-run80+**	**Cluster**
AT	-0.039	-0.010	0.001	0.000	1
BE	-0.039	-0.009	0.001	0.000	1
BG	-0.039	-0.010	0.001	0.000	1
CY	-0.039	-0.011	0.001	0.000	1
CZ	-0.039	-0.011	0.001	0.000	1
DE	-0.039	-0.003	0.001	0.000	4
DK	-0.039	-0.010	0.001	-0.000	1
EE	-0.039	-0.013	0.001	0.000	2
EL	-0.039	-0.014	0.001	0.000	2
ES	-0.039	-0.006	0.001	0.000	4
FI	-0.039	-0.005	0.001	0.000	4
FR	-0.039	-0.004	0.001	0.000	4
HR	-0.039	-0.010	0.001	0.000	1
HU	-0.039	-0.009	0.001	-0.000	1
IE	-0.039	-0.014	0.001	-0.000	2
IT	-0.039	-0.011	0.001	-0.000	1
LT	-0.039	-0.008	0.001	0.000	4
LU	-0.039	-0.019	0.001	-0.000	3
LV	-0.039	-0.012	0.001	0.000	1
MT	-0.039	-0.020	0.001	0.000	3
NL	-0.039	-0.006	0.001	0.000	4
PL	-0.039	-0.011	0.001	0.000	1
PT	-0.039	-0.010	0.001	0.000	1
RO	-0.039	-0.010	0.001	0.000	1
SE	-0.039	-0.012	0.001	0.000	1
SI	-0.039	-0.010	0.001	0.000	1
SK	-0.039	-0.008	0.001	0.000	4
Public expenditure on Pensions (p)					

Country	Short-run65+	Short-run80+	Long-run65+	Long-run80+	Cluster
AT	0.170	0.001	-0.023	0.006	1
BE	0.188	0.043	-0.023	0.002	2
BG	-0.019	-0.027	-0.023	-0.002	4
CY	0.265	0.005	-0.023	-0.007	2
CZ	-0.029	0.038	-0.023	-0.004	4
DE	0.131	0.004	-0.023	-0.000	1
DK	0.307	-0.056	-0.023	-0.002	3
EE	0.262	-0.027	-0.023	-0.007	2
EL	0.218	0.151	-0.023	0.004	2
ES	0.207	-0.053	-0.023	0.006	1
FI	0.298	-0.025	-0.023	0.000	3
FR	0.253	0.035	-0.023	0.006	2
HR	0.179	-0.028	-0.023	-0.005	1
HU	-0.027	0.208	-0.023	-0.009	4
IE	0.015	0.016	-0.023	-0.005	4
IT	0.225	0.019	-0.023	0.010	2
LT	0.306	-0.033	-0.023	-0.007	3
LU	0.119	-0.048	-0.023	-0.003	1
LV	0.216	-0.135	-0.023	-0.004	3
MT	0.150	-0.031	-0.023	-0.005	1
NL	0.185	-0.000	-0.023	-0.002	1
PL	0.212	0.067	-0.023	-0.000	2
PT	0.188	0.035	-0.023	0.005	2
RO	0.334	0.131	-0.023	-0.005	2
SE	0.174	-0.032	-0.023	-0.001	1
SI	0.130	-0.025	-0.023	-0.004	1
SK	0.125	0.051	-0.023	-0.007	1

Appendix D. Cross-sectional dependence diagnostics

This appendix reports the results of the Pesaran CD test applied to the residuals of the baseline ECMs. The test assesses whether unobserved shocks and expenditure dynamics remain correlated across EU countries after controlling for the included covariates and country fixed effects. The null hypothesis of cross-sectional independence is rejected for aggregate age-related expenditure, healthcare, and pensions. For long-term care (LTC), the CD statistic is not statistically significant. However, this result coexists with the highest mean absolute residual correlation among all expenditure categories (0.332), indicating the presence of substantial bilateral correlations that may offset each other in the aggregate test statistic. Taken together, the diagnostics suggest that residual cross-country dependence cannot be ruled out for LTC and that the pattern of dependence is likely more heterogeneous than in the other expenditure categories. Consequently, the same econometric treatment is retained across all specifications, ensuring methodological consistency and comparability of results.

Diagnostic	Age-related expenditure (a)	Healthcare (h)	Long-term care (ltc)	Pensions (p)
Pesaran CD statistic	5.688	6.454	-0.668	4.082
Two-sided p-value	<0.001	<0.001	0.504	<0.001
Mean absolute residual correlation	0.301	0.275	0.332	0.203

Note: The Pesaran CD test examines the null hypothesis of cross-sectional independence in the residuals of the baseline ECM. Two-sided p-values are reported. The mean absolute residual correlation summarises the magnitude of pairwise residual dependence across countries.

Appendix E. Country-level coverage of expenditure variables

This appendix documents the country-level availability and temporal coverage of the four expenditure variables used in the analysis. The table reports the number of non-missing annual observations and the firslast available years for each country and expenditure category, prior to constructing first differences and lagged variables for the estimations.

Country	Total age-related expenditure (a)		Healthcare (h)		Long-term care (LTC)		Pensions (p)	
	N	Period	N	Period	N	Period	N	Period
Austria (AT)	29	1995-2023	30	1995-2024	29	1995-2023	29	1995-2023
Belgium (BE)	21	2003-2023	21	2003-2023	21	2003-2023	29	1995-2023
Bulgaria (BG)	13	2011-2023	13	2011-2023	13	2011-2023	19	2005-2023
Croatia (HR)	11	2013-2023	11	2013-2023	11	2013-2023	16	2008-2023
Cyprus (CY)	14	2010-2023	14	2010-2023	14	2010-2023	24	2000-2023
Czechia (CZ)	14	2010-2023	15	2010-2024	14	2010-2023	29	1995-2023
Denmark (DK)	14	2010-2023	15	2010-2024	15	2010-2024	29	1995-2023
Estonia (EE)	21	2003-2023	21	2003-2023	21	2003-2023	25	1999-2023
Finland (FI)	24	2000-2023	24	2000-2023	24	2000-2023	32	1992-2023
France (FR)	18	2006-2023	19	2006-2024	18	2006-2023	32	1992-2023
Germany (DE)	32	1992-2023	32	1992-2023	32	1992-2023	32	1992-2023
Greece (EL)	15	2009-2023	15	2009-2023	15	2009-2023	29	1995-2023
Hungary (HU)	18	2006-2023	18	2006-2023	18	2006-2023	25	1999-2023
Ireland (IE)	13	2011-2023	14	2011-2024	14	2011-2024	29	1995-2023
Italy (IT)	12	2012-2023	13	2012-2024	13	2012-2024	29	1995-2023
Latvia (LV)	11	2013-2023	11	2013-2023	11	2013-2023	27	1997-2023
Lithuania (LT)	18	2006-2023	18	2006-2023	18	2006-2023	28	1996-2023
Luxembourg (LU)	13	2011-2023	14	2011-2024	14	2011-2024	29	1995-2023
Malta (MT)	10	2014-2023	10	2014-2023	10	2014-2023	29	1995-2023
Netherlands (NL)	26	1998-2023	27	1998-2024	27	1998-2024	29	1995-2023
Poland (PL)	11	2013-2023	11	2013-2023	11	2013-2023	24	2000-2023
Portugal (PT)	24	2000-2023	25	2000-2024	24	2000-2023	29	1995-2023
Romania (RO)	13	2011-2023	13	2011-2023	13	2011-2023	24	2000-2023
Slovakia (SK)	12	2012-2023	12	2012-2023	12	2012-2023	29	1995-2023
Slovenia (SI)	10	2014-2023	11	2014-2024	10	2014-2023	28	1996-2023
Spain (ES)	21	2003-2023	21	2003-2023	21	2003-2023	29	1995-2023
Sweden (SE)	13	2011-2023	13	2011-2023	13	2011-2023	31	1993-2023

Note: N denotes the number of non-missing annual observations. The reported period indicates the first and last available years but does not necessarily imply uninterrupted coverage between them. The effective estimation samples are further restricted by the construction of first differences and lags and by the joint availability of the control variables.